

Good Reasoning!

$\Delta H = i \cdot m \cdot K$
 $\pi = i M R T$
 $\frac{17.5}{20} \Rightarrow 88\%$

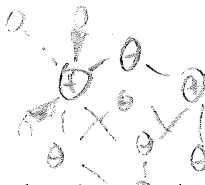
0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5	13	13.5	14	14.5	15	15.5	16	16.5	17	17.5	18	18.5	19	19.5	20
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Name: _____

These questions assess your understanding of the material from Chapters 10, 11 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. In an ionic crystal a cation (Cs^+ , radius 167 pm) sits among anions (Cl^- , radius 181 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

Simple cubic
Coordination number = 6



ANSWER: SC, 6

Radii are similar $\approx$
0.92:1

2. When a narrow wax (paraffin) capillary tube is dipped into water, does the water rise or become depressed in the tube? Justify using cohesive and adhesive forces.

Water - Paraffin attraction < water-water attraction

ANSWER: Depressed

Cohesion > Adhesion $\rightarrow$ water lowers

3. A monochromatic X-ray beam strikes a crystalline solid. Successive crystal planes produce a strong reflected beam under the conditions given below. Monochromatic X-rays of wavelength 179 pm produce the second-order diffraction maximum at a glancing angle of $\theta = 27.8^\circ$. Determine the spacing between adjacent crystal planes. Determine the requested quantity. A correct answer requires a justification based on the underlying physical cause rather than a memorized relationship.

Spacing must
be 202 pm, otherwise
the diffraction angle
would be shifted.

$$n\lambda = 2d \cos \theta$$

$$d = \frac{n\lambda}{2 \cos \theta}$$

2nd order
diffraction $\rightarrow n=2$

ANSWER: $2.02 \times 10^{-10} \text{ m}$
3SF

$$\frac{2(179 \times 10^{-12})}{2(0.8846)} \approx 2.02 \times 10^{-10} \text{ m}$$

4. For xenon (Xe): normal melting point -112°C , normal boiling point -108°C , triple point (-112°C , 0.807 atm), critical point (17°C , 58 atm). What phase is present at 90°C and 99 atm? Justify your answer.

Supercritical. Xe is above
Critical point temp & pressure

ANSWER: Supercritical

5. A solution shows a boiling point elevation of 0.186°C when 200. g of water is used as the solvent. How many grams of urea ($\text{CH}_4\text{N}_2\text{O}$, $M = 60.06 \text{ g/mol}$) are dissolved? ($K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$)

$$\Delta T_b = i \cdot m \cdot K_b$$

$$m = \frac{\Delta T_b}{i \cdot K_b} = \frac{0.186}{1 \cdot 0.512} = 0.3633 \text{ m}$$

ANSWER: 4.36g
3SF

or 4.364 g urea

6. The enthalpy of solution of ammonium nitrate (NH_4NO_3) is $\Delta H_{\text{sol}} = +25.7 \text{ kJ/mol}$. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when ammonium nitrate dissolves in water?

Positive enthalpy = energy absorbed
energy absorbed = endothermic = soln. cools.

ANSWER: Endothermic, decrease

7. An unknown solid has a melting point of 2852°C , is a non-conductor as a solid, and is a conductor when molten. Classify it as ionic, metallic, covalent-network, or molecular.

Metallic: always conductor ✓

Covalent: not conductors

Molecular: not usually conductors & low mp

Ionic: conductor when molten + high mp

ANSWER: Ionic ✓

8. Identify the dominant type of intermolecular force present in pure PH_3 . You must justify your answer.



Permanent Dipole + 3H donors & 2 acceptors

H-bonding can occur, but P is not as EN as N or F

ANSWER: Dipole-dipole ✓

9. When 6.72g of an unknown nonelectrolyte is dissolved in 200.g of water (H_2O), the freezing point decreases by 0.512°C . $K_f(\text{water}) = 1.86^{\circ}\text{C kg/mol}$. Calculate the molar mass of the unknown solute.

$$\Delta T_f = i \cdot m \cdot K$$

$$m = \frac{\Delta T_f}{i \cdot K} = \frac{0.512}{1 \cdot 1.86} = 0.2753 \frac{\text{mol}}{\text{kg}}$$

ANSWER: 24.4g/mol X

$$\frac{6.72\text{g}}{0.2753\text{mol}} = 24.41\text{g/mol} \quad \text{3SF}$$

10. Consider quartz (SiO_2). Predict whether this substance has a high or low melting point and whether it conducts electricity as a solid. Report both predictions and justify your answer. Responses based only on memorized trends or classification rules will not receive credit.

SiO_2 is a covalent-network solid, and with strong bonding, has a high mp.

SiO_2 is also covalent $\rightarrow$ insulating non-conductor

ANSWER: High, non-conducting ✓

11. How many grams of NH_3 can vaporize when 39.8kJ of heat is absorbed during vaporization at -33°C ? Assume $\Delta H_v = 1370\text{J/g}$.

$$\frac{39,800\text{J}}{1,370\text{J/g}} = 29.05\text{g}$$

ANSWER: 29.0g ✓

$$\frac{39.8\text{kJ}}{1.37\text{kJ/g}} = 29.0\text{g} \quad \text{3SF}$$

12. Rank the following equimolar aqueous solutions from smallest to largest boiling point elevation: NaCl , CaCl_2 , $\text{C}_6\text{H}_{12}\text{O}_6$. All solutions have the same molality.



larger bp elevation

ANSWER: $\text{C}_6\text{H}_{12}\text{O}_6$, NaCl , CaCl_2 ✓

13. Predict how the vapor pressure of water changes as its temperature increases. Justify your answer.

As temperature rises, more H_2O molecules escape the surface and become a gas, increasing vapor pressure.

ANSWER: Increases ✓

+6

14. Two solutions are prepared: solution A is a 0.200m solution of NaCl (NaCl) in water, and solution B is a 0.200m solution of glucose (C₆H₁₂O₆) in water. For water, $K_f = 1.86^\circ\text{C}\cdot\text{kg/mol}$ and $K_b = 0.512^\circ\text{C}\cdot\text{kg/mol}$. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.)

Soln B $\Delta H_b = i \cdot m \cdot K_b = 1 \cdot 0.2 \cdot 0.512 = 0.1024^\circ\text{C}$ ANSWER: Soln. A ✓
 Soln A $\Delta H_b = i \cdot m \cdot K_b = 2 \cdot 0.2 \cdot 0.512 = 0.2048^\circ\text{C}$

15. A 79.5-g sample of liquid water at 100°C absorbs 110.kJ of heat. Predict the equilibrium state after the heat transfer. Report the final temperature and account for the entire sample. If multiple phases are present, report the mass of each phase. Justify your answer. ($\Delta H_{\text{vap}} = 2260 \text{ J/g}$)

$\frac{110,000 \text{ J}}{2260 \text{ J/g}} = 48.67 \text{ g H}_2\text{O vapor}$ (3 sf.)
 $30.83 \text{ g H}_2\text{O liquid}$

ANSWER: Mixed vapor/liquid, 100°C

Insufficient heat to boil all water, temp stays at 100°C until all water boils.

16. List every type of intermolecular force present in pure PH₃. Justify your answer.

London-dispersion, dipole-dipole, and weak H-bonding

ANSWER: London-dispersion

No ions means no ion-dipole or ion-ion bonding.

NO FON

So H bonding is weak

17. Calculate the freezing point of a solution prepared by dissolving 10.0 g of NaCl (NaCl) in 100. g of water. $K_f(\text{H}_2\text{O}) = 1.86^\circ\text{C kg/mol}$. Assume complete dissociation ($i = 2$).

$\Delta H_f = i \cdot m \cdot K_f = 2 \cdot 1.71 \cdot 1.86 = 6.36^\circ\text{C}$

ANSWER: -6.36°C

$\frac{10 \text{ g NaCl}}{58.45} \approx 0.171 \text{ mol}$

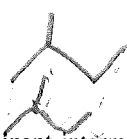
freezing point depression

18. Two substances have intermolecular potential-energy curves of identical shape but different well depths. Which forms the solid with the higher melting point? Justify.

Steeper well = more energy without shifting bonds = higher mp.

ANSWER: The one with the steeper energy well. ✓

19. Two isomers share the molecular formula C₅H₁₂: isopentane (branched) and n-pentane (straight-chain (elongated)). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.



VS



n-pentane has more surface area for London-dispersion attractions = higher bp

ANSWER: n-pentane ✓

20. Identify the dominant intermolecular force acting between potassium bromide (KBr) and water (H₂O) when they are mixed. Justify your answer.

KBr → ionic

H₂O → permanent dipole

KBr · (H₂O)_x → ion-dipole

ANSWER: Ion-dipole

K⁺ and Br⁻ are attracted to the dipoles in H₂O, and form ion-dipole attractions.

+ 6.5

1. Cubic holes, coordination number 8 $\therefore r_+/r_- = 167/181 = 0.92$ lies in the 0.732–1.0 range, which geometrically favors cubic coordination.
2. It will be depressed (fall) $\therefore$ adhesion to wax (paraffin) is weaker than cohesion, so the liquid is pushed down.
3. $d = \frac{n\lambda}{2\sin\theta} = \frac{2 \times 179}{2\sin(27.8^\circ)} = 384 \text{ pm}$. $\therefore$ waves reflected from adjacent crystal planes travel different distances. A diffraction maximum occurs only when that path difference equals an integer multiple of the wavelength, so constructive interference requires $n\lambda = 2d\sin\theta$.
4. Supercritical fluid $\therefore$ both T and P exceed the critical point (17°C , 58 atm).
5. $m = \frac{\Delta T}{K_b} = \frac{0.186}{0.512} = 0.363 \text{ mol/kg}$; $n = 0.363 \times 0.200 \text{ kg} = 0.0727 \text{ mol}$; $m_{\text{urea}} = 0.0727 \text{ mol} \times 60.06 \text{ g/mol} = 4.36 \text{ g}$
6. Endothermic ($\Delta H_{\text{sol}} = 25.7 \text{ kJ/mol}$) $\therefore$ solution temperature will decrease. $\therefore$ lattice energy > solvation energy; system absorbs heat; entropy increase drives dissolution.
7. Ionic solid (MgO) $\therefore$ ionic crystal with high melting point due to $+2/-2$ charges; non-conducting as solid.
8. Dipole-dipole $\therefore$ pyramidal polar molecule; P–H does not form H-bonds.
9. $m = \frac{\Delta T_f}{K_f} = \frac{0.512}{1.86} = 0.275 \text{ mol/kg}$; $n = m \times m_{\text{solvent}} = 0.275 \times 0.200 \text{ kg} = 0.0551 \text{ mol}$; $\mathcal{M} = \frac{6.72 \text{ g}}{0.0551 \text{ mol}} = 122 \text{ g/mol}$
10. High melting point: Yes. Conducts as a solid: No. $\therefore$ Its atoms are connected by covalent bonds throughout an extended three-dimensional network. Breaking the solid requires breaking many covalent bonds and there are no mobile charged particles.
11. $m = \frac{q}{\Delta H} = \frac{39.8 \times 10^3 \text{ J}}{1370 \text{ J/g}} = 29.1 \text{ g}$
12. $\text{C}_6\text{H}_{12}\text{O}_6$ ($i = 1$) < NaCl ($i = 2$) < CaCl_2 ($i = 3$) $\therefore$ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. $\text{C}_6\text{H}_{12}\text{O}_6$: nonelectrolyte; 1 particle per formula unit; NaCl : strong electrolyte; $\text{Na}^+ + \text{Cl}^- \rightarrow 2$ particles; CaCl_2 : strong electrolyte; $\text{Ca}^{2+} + 2\text{Cl}^- \rightarrow 3$ particles.
13. Vapor pressure increases $\therefore$ a larger fraction of molecules gains enough kinetic energy to escape the liquid, so more enter the vapor before equilibrium is re-established.
14. Solution A: $\Delta T = iK_m = 2 \times 0.512 \times 0.200 = 0.205^\circ\text{C}$; Solution B: $\Delta T = 1 \times 0.512 \times 0.200 = 0.102^\circ\text{C}$. Solution A has the greater boiling point elevation $\therefore$ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).
15. Vaporizing all 79.5 g requires $79.5 \times 2260 = 180. \text{ kJ}$, but only 110. kJ is supplied $\therefore$ a liquid water–steam mixture at 100°C . Liquid water = 30.8 g; steam = 48.7 g. $\therefore$ the supplied energy is insufficient to vaporize the entire sample, the absorbed energy is used for the phase change rather than raising the temperature, so both phases remain at the boiling point.
16. London dispersion and dipole–dipole $\therefore$ weakly polar; P–H does not hydrogen-bond.
17. $m = \frac{0.171 \text{ mol}}{0.100 \text{ kg}} = 1.71 \text{ mol/kg}$; $\Delta T_f = iK_f m = 2 \times 1.86 \times 1.71 = 6.37^\circ\text{C}$; $T_f = 0.00 - 6.37 = -6.37^\circ\text{C}$
18. The substance with the deeper well $\therefore$ a deeper well means stronger attractions, so more thermal energy is required to disrupt its lattice and melt it.
19. N-pentane $\therefore$ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.
20. Ion-dipole forces $\therefore$ the K^+ and Br^- ions attract the partial charges of polar water.